

A metal price forecasting framework optimized with TLBO metaheuristic-algorithm and selective for highest returns

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Abstract

Metal price prediction constitutes a significant economic challenge. Fluctuations in metal prices substantially impact various industries, including manufacturing, construction, energy, and transportation. Accurate price predictions enable organizations to make informed decisions, optimize operations, and mitigate risks, thereby enhancing competitiveness and profitability. The proposed framework is designed to process time series price data with the target output including trend predictions and recommended actions for each day to achieve the highest return. In this study, we investigate various machine-learning algorithms, including linear regression, logistic regression, Artificial Neural Networks, Long Short-Term Memory, and XGBoost, for both classification and regression output. This research employs Empirical Mode Decomposition to generate decomposed series, termed Intrinsic Mode Functions (IMFs), from close price time series data. Subsequently, these IMFs are utilized as feature vectors for prediction models. To optimize returns in trading decisions, the study applies the TLBO algorithm to determine the optimized learning rate, training and validation datasets, and SHRV method to select the models that achieve the highest return at validation rather than the lowest cross-entropy or MSE loss for the optimized models. In the experiment involving seven different types of metal, all classification-based models using IMFs as input features achieved an average annual return exceeding 273.96%. Among these results, the ANN classification optimized by TLBO and SHRV based methods demonstrated the highest performance with an average annual return of 351.98%, a daily return of 0.5496, and an accuracy of 70.01%. The model exhibited the best performance for silver, with the balance increasing significantly by 1553 times after 995 trading days.

Keywords: metal price forecasting, model optimization, return optimization, machine learning, hybrid models, empirical mode decomposition, time series prediction

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1. Introduction

Research on metal price forecasting, specifically commodity and stock price forecasting in general, commenced decades ago to address the demand for forecasting services. However, the continuous pursuit of higher profits and more accurate forecasting models remains a key driver of innovation in this field. This has motivated researchers to persistently endeavor to create superior price prediction models by applying the most advanced achievements in the field of computer science. In this research project, we aim to utilize state-of-the-art advancements in the field of machine learning to design a novel hybrid model to predict the prices of important metals, such as copper, aluminum, nickel, zinc, iron, silver, and gold in the futures market. We intend to focus on addressing the limitations of previous studies in the field of metal price forecasting, such as the lack of actionable buying recommendations. The majority of the results only evaluated the effectiveness of the model based on metrics such as the MAPE, MSE, and RMSE for price or volatility. It can be challenging for an individual to make a trading decision after receiving the optimal price movement prediction. For instance, if the model predicts the price of copper for the next month to be 3.90 USD per pound, what specific actions should company managers take to maximize benefits? Should they purchase because the price will be higher than the current price, or wait because the price may decrease further? These recommended actions are not provided by those models. To address this issue, our research focuses on utilizing the profit return of prediction models as the primary metric, in addition to addressing the challenges of the price forecasting field, such as data quality, data harmonization, nonlinear processing, market sentiment analysis, incorporating external factors, transparency and interpretability, continuous monitoring, and model updating. We propose a framework for predicting metal prices in the futures market, based on a hybrid approach.

The study of metal price predictions in futures markets holds significant importance for economic and industrial management. Non-ferrous metals constitute essential industrial materials that provide strategic support for national economic development. Price forecasting for non-ferrous metals is crucial for investors, policymakers, and researchers [1]. For instance, copper is extensively utilized in various industries, including machine manufacturing, automobile production, agriculture, electronic components, household appliances, musical instruments, and commemorative monuments [2].

Metal prices are the result of complex market dynamics and stochastic economic processes that render price forecasting challenging [3]. Metal markets are influenced by numerous factors, including geopolitical events, supply-demand dynamics, and economic policies. Research on metal price prediction can contribute valuable insights to the field, potentially leading to novel methodologies, enhanced models, and a more comprehensive understanding of market dynamics. Investigations in this domain can inform policymakers and industry leaders by

enabling them to make evidence-based decisions that affect economic growth, infrastructure development, and resource management. Accurate metal price predictions can contribute to market stability and mitigate economic shocks, benefiting not only investors and traders, but also consumers and enterprises dependent on stable metal supplies.

Metal price volatility poses substantial risks to industries that rely heavily on these materials. Accurate price predictions of important metals such as copper, aluminum, nickel, zinc, and iron enable industrial managers to make informed decisions regarding procurement, production planning, and budgeting. Considerable evidence suggests that cyclical behavior is a dominant characteristic of metal prices [3]. Labys et al. [4] found short-term cycles of durations of less than 12 months in monthly prices of aluminum, copper, lead, zinc, and other metals and encouraged producers, consumers, and traders to re-examine their price forecasting methods based on this finding [3]. Copper will not become physically scarce in the future; however, increased demand and decreased resource quality will result in significant price increases. It has been posited that the price of copper will increase significantly in the coming decades. A similar situation applies to zinc and lead, with soft scarcity and increased prices for zinc [5]. Understanding future price trends can inform decisions regarding the timing of raw material purchases, inventory management, and production scaling. By predicting these prices, managers can better plan and control costs, leading to optimized profit margins. Predictive research aids in identifying potential price risks and developing strategies to mitigate them, such as hedging or diversifying suppliers. In industries where metals are key inputs, future price predictions can inform supply chain strategies. This includes negotiating contracts with suppliers, determining long- and short-term purchase strategies, and managing the risks associated with price volatility.

Price forecasting of raw materials in the real world is a complex task in which multiple factors are interconnected and correlated. Non-ferrous metals are raw materials with both commodity and financial attributes [1]. This research area presents an opportunity to work with complex datasets and employ advanced analytical techniques, including machine learning and artificial intelligence. Liu et al. [1] presents a novel hybrid deep learning model, which combines the VMD (variational mode decomposition) method and the LSTM (long short-term memory) network to construct a forecasting model. Zhang et al. [6] develops two novel hybrid intelligence models for forecasting copper prices in the future with high accuracy based on the extreme learning machine (ELM) and two meta-heuristic algorithms. Liu et al. [7] proposed a model using a decision tree to predict future copper prices in the long and short term. Du et al. [8] designs an innovative hybrid model using a robust decomposition method, machine learning, and nature-inspired metaheuristic technique for multi-step copper price prediction. Khoshalan et al. [9] proposed AI methods, Gene expression programming (GEP), artificial neural network (ANN), Adaptive neuro-fuzzy inference system (ANFIS), and ANFIS – ACO for forecasting the price of copper metal, utilizing coal, aluminum, crude oil, gold, iron ore, natural gas, nickel, and lead as selected parameters. Zhou et al. [10] proposed RSAR model con-

sists of a convolutional neural network (CNN) and Long Short-Term Memory Neural Networks (LSTM), integrating with self-attention mechanism to extract both spatial and temporal features from precious metal price data. Zhang et al. [2] develop and propose several forecast models for predicting monthly copper prices in the future based on various algorithms in machine learning, including multi-layer perceptron (MLP) neural network, k-nearest neighbors (KNN), support vector machine (SVM), gradient boosting tree (GBT), and random forest (RF). Zhao et al. [11] develop and propose a novel model based on an ensemble machine learning technique to predict monthly copper prices in the future by using different soft computing methods, including multi-layer perception (MLP) neural network, support vector regression (SVR), and extreme gradient boosting (XGBoost). Ozdemir et al. [12] propose LSTM and GRU networks to forecast the nickel price in the medium- and long-term horizon. Although numerous machine-learning-based methods have been employed, there remain significant limitations in the published literature and future work in the field of metal price prediction. The rapid advancement of artificial intelligence offers the potential to create models that demonstrate improved performance and generate increased profits.

The main contributions of this study are summarized as follows.

- (1). To the best of our knowledge, this paper is the first study to evaluate the profit return of a metal price prediction model.
- (2). The study demonstrates that the TLBO algorithm identifies the optimized learning rate and ratio of the training and validation dataset to enhance the performance of the classification-based price-prediction models.
- (3). This study applies and evaluates a novel methodology for the optimized model in the training and validation phases by selecting models that achieve the highest return at validation (SHRV) rather than the lowest cross-entropy or MSE loss.
- (4). In the study, the ANN classification algorithm optimized by the TLBO algorithm and SHRV demonstrated superior performance, achieving an average annual return of 351.98%, daily return of 0.5496%, and accuracy of 70.01% for seven metals, including copper, aluminum, nickel, zinc, iron, silver, and gold.
- (5). The proposed method demonstrates high accuracy in predicting significant price movements, which is an important aspect of price prediction.

The remainder of this paper is organized as follows. Section 2 discusses literature review. Section 3 describes the theoretical background and methodology used in this study. Section 4 presents the experiment results and analysis. Section 5 presents conclusion and future works.

2. Literature review

2.1. Econometric Models

In the field of price prediction, we can separate models into three categories: econometric, machine learning, and AI models, as well as integrative and hybrid models, which represent different approaches for data analysis and forecasting,

each with its own methods and applications. Each of these modeling approaches has advantages and is suitable for different types of problems. The choice of model depends on the specific requirements of the task, nature of the available data, and objectives of the analysis. In particular, integrative and hybrid models are becoming increasingly popular because they offer a way to combine the explanatory power of econometric models with the predictive accuracy and flexibility of machine learning and AI technology.

2.2. Machine Learning and AI Models

These models originate from the economic theory and are used to describe economic and financial phenomena. They often use statistical techniques to test hypotheses and forecast future trends. Econometric models often rely on historical data, and assume that past patterns will continue in the future. They often use regression analysis and rely on the assumption that the relationship between the variables is linear and stable over time. Commonly used in economics and finance to forecast economic indicators, evaluate the impact of policy changes, and understand the relationships between different economic factors.

Dooley et al. [13] investigates the efficacy of time series forecasting techniques in predicting future prices of lead and zinc. This study focuses on two methods: a lagged forward price model and an Autoregressive Integrated Moving Average (ARIMA) model. This study analyzes two methods: lagged forward price modeling and ARIMA. The lagged forward price model uses past prices to predict future prices, whereas the ARIMA models use past price movements to forecast future trends. ARIMA modeling showed marginally better forecast results than lagged forward price modeling for the data analyzed. The findings suggest that although perfect forecasting accuracy is not achievable, these methods can still be valuable for mining companies. They provide a relatively simple and accessible method to conduct in-house analyses for planning and risk management.

Kriechbaumer et al. [3] proposed an advanced method for forecasting the monthly prices of metals like aluminium (alu), copper, lead, and zinc. The study introduces an improved forecasting approach that combines wavelet analysis with the autoregressive integrated moving average (ARIMA) model. This combined method identifies the optimal combination of wavelet transform type, wavelet function, and the number of decomposition levels used in the multiresolution analysis (MRA). By integrating wavelet-based MRA with ARIMA model fitting, the approach significantly enhances forecast accuracy. Despite achieving high forecast accuracies for most periods in the metal price series analyzed, the calibrated wavelet-ARIMA model encountered challenges during periods of sudden price changes.

Chen et al. [14] present a method for forecasting metal prices using a grey wave forecasting model. This method is developed to address the challenges posed by the severe fluctuations and irregular cycles in metal prices, making accurate forecasting difficult. This novel method employs unequal-interval contour lines based on the quantiles of the data. The novel method showed improved forecasting accuracy compared to basic grey wave forecasting and outperformed

traditional models like Random Walk (RW) and ARMA in multi-step-ahead prediction. For limitations, the grey wave forecasting method is a graphical prediction method and relies heavily on the appearance of the time series graph. This approach may overlook other characteristics of the time series, such as long-term memory.

Sverdrup et al. [5] focus on assessing the long-term availability and sustainability of copper, zinc, and lead resources. This study used the WORLD6 system dynamics model to simulate the supply and demand dynamics of these metals. While the model aims to capture the dynamics of metal supply and demand, it may not fully encompass all real-world complexities and interdependencies, especially the unforeseen economic, technological, and policy changes that can affect metal markets. System dynamics models are better at understanding and explaining past trends than at predicting future occurrences. Therefore, although they can model potential future scenarios, the accuracy of these forecasts can be limited, particularly in the face of unexpected global changes. External factors, such as technological advancements, the discovery of new resources, and significant policy shifts, can impact metal supply and demand.

2.3. Integrative and Hybrid Models

These models are a subset of artificial intelligence that focuses on developing algorithms that can learn and make predictions or decisions based on data. Machine-learning models are data-driven and do not necessarily require a pre-defined theoretical model. They are capable of handling nonlinear relationships and can adapt to changing patterns in data. These models cover a wide range of techniques from simple linear regression to complex neural networks. It is used in a wide range of fields including image and speech recognition, natural language processing, medical diagnosis, and stock market trading.

Ozdemir et al. [12] proposed a method to forecast the price of nickel using advanced deep learning techniques. This study aims to address the uncertainty in nickel price trends, which significantly impacts the mining investment, mine planning, and economic development of companies and countries relying on nickel resources. LSTM and GRU networks were found to be effective in forecasting nickel price fluctuations. The GRU network outperformed the LSTM network in terms of the computational efficiency.

Liu et al. [7] proposed a model using a decision tree to predict future copper prices in the long term and the short term. Pearson's correlation coefficients were used to analyze seven other commodity prices as affecting factors: crude oil, natural gas, gold, silver, lean hogs, coffee, and the Dow Jones index. The results showed that copper prices were highly correlated with crude oil, silver, coffee, gold, and lean hogs, whereas they were weakly correlated with the Dow Jones Index and Natural gas. The past seven factors and copper prices were used as feature vectors to predict the future copper prices.

Liu et al. [7] proposed a model using LSTM and GRU to train and predict the price of copper. The closing price was normalized to the minimum and maximum prices. It was then denoised, and irrelevant information was removed using a Wavelet Transform. The Bayesian Optimization algorithm was used

to determine the optimized hyperparameters after 12 iterations. The model achieved a prediction RMSE lower than 3% for three different periods: 100, 150, and 200 days. GRU performed better than the LSTM in the experiment.

2.4. Integrative and Hybrid Models

These models combine elements from different modeling approaches, often integrating traditional statistical or econometric methods with machine-learning techniques. Hybrid models aim to leverage the strengths of various approaches to improve accuracy and robustness. For example, a model may use econometric methods to capture long-term trends while employing machine learning algorithms to adapt to short-term fluctuations. Common in scenarios where complex phenomena are modeled, such as financial market forecasting, climate modeling, and advanced analytics in business intelligence.

Zhang et al. [6], two hybrid models were proposed: GA-ELM and PSO-ELM. ELM is an efficient training algorithm for artificial neural networks with a single hidden layer. This approach ensures a rapid training phase with high generalizability. In these models, the GA and PSO were used to optimize the weights of the ELM neural network. This optimization aims to improve the accuracy of the ELM model for forecasting copper prices. GA is an evolutionary algorithm used to find optimal solutions by mimicking natural selection processes. It includes the stages of selection, crossover, and mutation to search for the most suitable solution in a large search space. PSO is a computational method inspired by the social behavior of flocks (such as birds or fish) and is used to solve optimization problems. It involves particles flying around in the search space and continuously updating their velocities based on shared experience to achieve better positions. This study uses time-series data on copper prices from 1990 to 2019, along with other relevant variables such as crude oil, iron ore, gold, silver, natural gas prices, and foreign exchange rates. Statistical tests, such as the t-test and Wilcoxon test, were conducted to evaluate the significance of the proposed models.

Khoshalan et al. [9] proposed different artificial intelligence methods to predict copper prices. Specific algorithms used in the article include Gene Expression Programming (GEP), Artificial Neural Network (ANN), Adaptive Neuro Fuzzy Inference System (ANFIS), and ANFIS combined with algorithms Ant Colony Optimization (ANFIS-ACO). The paper used eight features that they identified as influencing copper prices, including the prices of coal, aluminum, crude oil, gold, iron ore, natural gas, nickel, and silver. This integrative method uses a variety of artificial intelligence and machine-learning models to predict copper prices. Each of these methods represents an aspect of machine learning or AI. GEP is an evolutionary algorithm that helps to develop computer programs to model complex relationships. ANN mimic the functioning of the human brain and are an important part of machine learning for pattern recognition and prediction. ANFIS integrates fuzzy logic with neural networks to handle uncertainty and imprecision, and the ANFIS-ACO hybrid model further enhances this approach using optimization techniques inspired by natural

processes. This approach differs from traditional econometric models, which often rely on established statistical methods and economic theories. Instead, the proposed methodology is more in line with modern AI and machine learning techniques, with an emphasis on data-driven model development, adaptability, and complex pattern recognition. The integrated nature of using multiple AI and machine learning techniques allows for a more robust and comprehensive analysis, potentially leading to more accurate copper price forecasting.

Du et al. [8] combined data preprocessing techniques, machine learning algorithms, and optimization strategies to enhance the accuracy of metal price forecasting. The use of a robust data-processing algorithm, specifically time-varying filter-based empirical mode decomposition (TVF-EMD), decomposes the time series into several modes. This helps in effectively identifying and extracting the main characteristics of the metal price time series. The extreme learning machine is optimized using nature-inspired metaheuristic algorithms, particularly the marine predator algorithm (MPA). This optimization enhances the prediction performance of the ELM. The combination of the aforementioned techniques into a hybrid model allows for more accurate multistep forecasting of copper and gold prices. This integrative approach leverages the strengths of different methods, such as forecasting models, optimization algorithms, and decomposition methods, to improve the overall forecasting capability.

Liu et al. [1] combine Variational Mode Decomposition (VMD), a signal processing technique, and Long Short-Term Memory (LSTM) networks, an advanced machine learning method, to create a comprehensive forecasting model. The authors address the challenge of predicting these prices owing to their significant fluctuations and irregular cycles, which makes accurate and robust forecasting difficult. VMD was employed to disassemble the original price series into several components, each representing different aspects of price movements. The LSTM network was then used to separately forecast each component. Finally, the forecasts for all components were aggregated to create a comprehensive prediction for the original price series. Extensive experiments were conducted using the daily futures prices of Zinc, Copper, and Aluminum from the London Metal Exchange (LME). The study concludes that the VMD-LSTM model, with its ability to capture the internal characteristics of the data and learn the interdependence of historical data, effectively improves the overall prediction ability and is a promising methodology for various forecasting fields. The paper notes that the real-time demand for price forecasting in practical applications is not high; therefore, this computational cost may not be a significant issue. In addition, with advancements in hardware computing power and the application of parallel computing technology, the runtime of the proposed method is expected to decrease in the future.

Wang et al. [15] combine a price volatility network (PVN) and 3 ANN techniques including BPNN, RBFNN, and ELM to forecast future copper prices. The hybrid PVN-ANN techniques revealed nonlinear patterns of prices to predict both the level and direction. The model performed experiments with daily, weekly, and monthly datasets, and achieved average MAPE values of 1.55%, 3.41%, and 5.29%, respectively. The PVN-ANN technique achieved better per-

formance than the traditional ANN techniques. The Friedman test was used as a statistical method to analyze the MAPE results and calculate the ranking for each predictive technique.

Some machine learning-based and hybrid models used for stock and commodity forecasting can also be considered when designing metal price prediction models. Zhao et al. [16] proposed a method using LSTM to process the time series data of the stock market's prices combined with the timeliness of emotion. This model was used to predict the closing price of an HSI using data from the previous three years. A denoising autoencoder was used to extract key features from 6 features of stock prices. An emotion-enhanced convolutional neural network (ECNN) was used to calculate sentiment indices from textual content [17] collected from the Internet to reflect investor sentiment. The study also proposed some metrics to evaluate the performance, profit, and risk of the models, including the Mean Absolute Percentage Error (MAPE), annual return (AR), Sharp Ratio (SR), and annual volatility (V).

Md et al. [18] proposed the multi-layered sequential LSTM model using some vanilla LSTM layers and a Dense layer to process the time series of copper close price. The time-series data were normalized using a min max scaler and divided into time steps of 100 data points each. The model used the MSE loss function and was compared with RNN, LSTM, CNN, and MLP.

Celik et al. [19] proposed a hybrid model using ANN, RF, XGBoost, EMD, and LIME algorithms to predict the direction of six stock market indices. The paper claims that the two drawbacks of most machine-learning-based models are their black-box nature and unexplainable predictions. This study proposes the LIME algorithm as an explainable AI approach to evaluate the reliability of each prediction. The EMD algorithm was applied to reduce the complexity of the original time series by decomposing it into more manageable subcomponents called the IMF. Each IMF can represent different time-scale components of the original data. The model applied ANN in the regression and classification modes to calculate the two prediction results, and RF and XGBoost were used to predict the direction class of the scaled original price series. This study attempted to integrate the LIME algorithm to determine the reliability level for each prediction and applied predictions with higher reliability to make decisions. The experiment showed that RF performed better than XGBoost did. Increasing the reliability levels led to an increase in accuracy but a decrease in the number of trusted predictions.

Wang et al. [20] apply a transformer model to predict the stock market index using daily closing price data. A moving-window approach is used to create the input features of the model to predict the next data. The author claimed that their transformer model had some limitations in using only a single stock market index without using a high-dimensional time series. This study evaluates performance using profit return, MAPE, MSE, Volatility, and Sharpe ratio. The transformer model achieved the best returns using a simple trading strategy. Zhang et al. [21] proposed a framework called TEANet which used Transformer Multihead Attention, Temporal attention, and LSTM to predict the stock movement. The model designed feature engineering to process

text from social media platforms using a Transformer Encoder and then fused text-based features with price-based features using a concatenation processor. LSTM and Temporal attention mechanisms are used to capture the temporal dependencies of the fused features.

3. Theoretical background and methodology

3.1. Machine learning-based method

The analysis and prediction of time series data utilizing machine learning (ML) methodologies has become indispensable in diverse domains such as finance, healthcare, economics, and meteorology. This approach endeavors to extract insights and generate forecasts based on temporally ordered information. Time series data exhibits several distinctive characteristics, including temporal dependencies, trends and cyclical patterns, stochastic fluctuations, and non-uniform intervals between data points.

Various conventional models can be employed to predict stock and metal prices, including ARIMA and SARIMA. ARIMA is widely utilized for forecasting stationary time series by modeling the relationships between lagged observations. SARIMA extends ARIMA by incorporating seasonality. Furthermore, fundamental machine learning techniques such as linear regression, logistic regression, and decision trees can be applied. Linear regression assumes a linear relationship between input (historical price data) and output (future price). While it can serve as a foundational model for stock price prediction, it is generally insufficiently complex to capture the intricate patterns in stock price fluctuations. Logistic regression can categorize the direction of stock price movement (upward or downward) rather than forecasting the actual price. Decision Trees are a non-parametric supervised learning approach used for both classification and regression tasks. Their objective is to develop a model that predicts the value of a target variable by learning simple decision rules derived from data features. In stock price forecasting, Decision Trees can be employed to both predict future stock prices (regression trees) and classify the direction of stock price movement (classification trees).

Various Deep Learning Models encompass ANN, LSTM, GRU, and CNN. Artificial Neural Networks (ANN) possess the capability to process substantial volumes of input data to identify complex patterns and non-linear relationships within the information. This attribute renders ANNs particularly valuable for modeling the inherently unpredictable and non-linear stock market. Long Short-Term Memory networks (LSTMs), a variant of recurrent neural networks (RNNs), are specifically designed for sequence prediction tasks. Given the sequential nature of stock data, LSTMs can effectively forecast future stock prices by learning from extensive historical data sequences. They demonstrate proficiency in recognizing temporal dependencies and patterns across different time frames, making them highly effective for prediction tasks where past price trends may indicate future movements. Gated Recurrent Units (GRU), another type of RNN, are particularly adept at handling sequential data such as time

series. Similar to other RNNs, GRUs utilize internal memory to process input sequences but are less complex than LSTMs, employing fewer parameters. This results in reduced training times while often achieving comparable performance. Convolutional Neural Networks (CNNs), traditionally employed for image processing, are also effective for any data with spatial structure, including time series. Their capacity to perform convolution operations allows them to filter inputs for pertinent information, enabling the identification of patterns and features in data.

Random forests and XGBoost are exemplars of ensemble techniques. Random Forest is a machine learning methodology that constructs multiple decision trees during training and generates outputs based on the majority vote (for classification) or mean prediction (for regression) of these trees. XGBoost, conversely, is a high-performance implementation of gradient-boosted decision trees, extensively employed in regression and classification tasks. In the context of stock price prediction, XGBoost's versatility in handling heterogeneous data types and its robust predictive capabilities render it well-suited for modeling price fluctuations with precision. XGBoost's capacity to automatically manage missing data and maintain accuracy even with imperfect training data, which is prevalent in financial datasets, is particularly advantageous. Furthermore, certain advanced hybrid models combine diverse approaches, such as integrating ARIMA with LSTMs or RNNs, to leverage the strengths of each method for enhanced prediction accuracy.

In conclusion, while machine learning provides robust methodologies for metal price prediction, it is imperative to select algorithms that are appropriate for the specific characteristics of the data and the forecasting objectives. As research and innovation in this domain continue to advance, it is anticipated that increasingly sophisticated methods will be developed, potentially enhancing the accuracy and reliability of stock market predictions.

3.2. Empirical Mode Decomposition

Empirical Mode Decomposition (EMD) is a sophisticated signal processing technique that has gained prominence for analyzing non-stationary and nonlinear signals. EMD serves as an adaptive, data-driven method that decomposes a signal into its intrinsic mode functions (IMFs) based on the local characteristics of the data. The fundamental principle of EMD involves the extraction of IMFs through a process of sifting, which iteratively identifies the local extrema of the signal. The method begins by locating the local maxima and minima, and then constructing upper and lower envelopes using spline interpolation. The mean of these envelopes is computed and subtracted from the original signal to yield the first IMF. This process is repeated on the residual signal until a stopping criterion is met, typically when the residual becomes a monotonic function [22].

EMD's versatility permits integration with other methods, such as machine learning, enhancing predictive modeling and forecasting accuracy [23]. Moreover, EMD has been shown to be particularly effective in scenarios where traditional methods, such as Fourier Transform or Wavelet Transform, may fall short. For instance, while Fourier Transform assumes stationarity and linearity,

EMD's adaptive nature allows it to handle non-stationary signals without the need for prior assumptions about the signal's characteristics [24]. This adaptability makes EMD a powerful tool for analyzing complex time series data across various domains.

Empirical Mode Decomposition is a robust and flexible technique for analyzing nonlinear and non-stationary signals. Its capacity to decompose signals into intrinsic mode functions without relying on predefined basis functions distinguishes it from traditional methods. As research continues to progress, EMD is likely to find increasingly diverse applications across various fields, further establishing its position as a critical tool in signal processing. Çelik et al. [19] applied Empirical Mode Decomposition (EMD) to decompose stock price data into Intrinsic Mode Functions (IMFs). Subsequently, they implemented two distinct Artificial Neural Network (ANN) models to forecast each IMF independently. The results demonstrate that EMD effectively reduced the complexity of the original data and enhanced the accuracy of the ANN models by providing more refined and structured input data. Zhou et al. [25] employed CEEMD, an enhanced version of EMD to enhance predictability by separating various frequency components and minimizing noise. By adding white noise pairs, the CEEMD technique enhances performance and addresses issues such as audio blending that are commonly encountered in conventional EMD. The process involves breaking down the historical precious metal price series into multiple IMFs and a residual component. Following this decomposition, the resulting subseries are input into an Optimized Relevance Vector Machine (RVM) model to generate initial predictions.

3.3. Teaching-Learning-Based Optimization algorithm

Teaching-Learning-Based Optimization (TLBO) is a novel algorithm inspired by classroom educational processes, introduced by Rao et al. in 2011. It has rapidly emerged as a potent tool for addressing diverse optimization problems in fields such as engineering, computer science, and operations research [26]. The fundamental principle of TLBO is the interaction between teachers and learners, wherein teachers impart knowledge and learners enhance their understanding and performance through this transfer [27]. This approach effectively balances exploration and exploitation, rendering TLBO a robust alternative to conventional techniques such as genetic algorithms (GA) and particle swarm optimization (PSO) [28].

A significant advantage of TLBO is its parameter-free nature, which simplifies implementation compared to metaheuristic algorithms that require extensive parameter tuning [27]. In TLBO, the sole parameter to define is the teaching factor (TF), which determines the extent of knowledge transfer from the teacher to the learners [29]. This characteristic enhances usability and robustness, enabling adaptation to various optimization landscapes without extensive parameter optimization [28]. The algorithm comprises two primary phases: the teacher phase, in which the optimal solution (teacher) guides the learners, and the learner phase, in which learners refine their solutions based on their experiences and the teacher's knowledge. The TLBO algorithm has been applied to

various optimization problems, including constrained and unconstrained tasks, multi-objective optimization, and engineering challenges. It has demonstrated superior performance in mechanical design optimization compared to traditional methods [30]. Research indicates that TLBO surpasses numerous optimization algorithms in terms of speed and solution quality, rendering it a preferred choice among researchers and practitioners [31]. Extensive benchmarking against standard test functions has validated its performance, demonstrating its efficiency in high-dimensional optimization problems [28].

Furthermore, TLBO's simplicity and ease of implementation have facilitated its integration into various software tools and frameworks, supporting both academic and industrial applications [32]. Teaching-Learning-Based Optimization (TLBO) represents a significant advancement in optimization algorithms, drawing inspiration from classroom educational dynamics to enhance knowledge transfer and solution quality. Its parameter-free design, rapid convergence, and broad applicability render it an effective tool for diverse optimization problems. Ongoing research on TLBO holds promise for further advancements in optimization techniques.

The Teaching-Learning-Based Optimization (TLBO) algorithm operates by allowing solutions, referred to as learners, to enhance themselves through the guidance of a teacher and interactions with peers. The performance of each hyperparameter configuration is evaluated based on its ability to successfully train a model. The main goal is to minimize validation loss to the lowest achievable value. In our approach, the TLBO process is divided into three distinct phases.

During the initial stage, the concept involves the top-performing individual in the population (designated as the teacher) attempting to guide other learners (solutions) towards an optimal solution by altering the population's mean. The approach employs the teacher formula 1, where X_i^{new} represents the updated individual's parameter, X_i denotes the current learner being modified, X_T signifies the teacher with the best solution in the population, X_{mean} is the population's current average, T_f represents the teaching factor calculated by rounding $(1 + \text{rand}(0,1))$, and r is a random value within the range $[-1, 1]$.

$$X_i^{new} = X_i + r * (X_T - T_f * X_{mean}) \quad (1)$$

During the second phase, individual learners enhance their performance through interactions with one another. For each learner X_i , a learner X_j is randomly chosen. If X_i has a better objective function value than X_j , it learns from itself, otherwise, it learns from X_j . The model apply the learner formula 2, where X_i^{new} represents the updated individual's parameter, X_i denotes the current learner, X_j is the randomly selected other learner, and r signifies a random value within the range $[-1, 1]$. $f(x)$ represents the evaluation function, wherein the output is the loss value. Consequently, the objective of TLBO is to identify the solution X_i that minimizes $f(x)$.

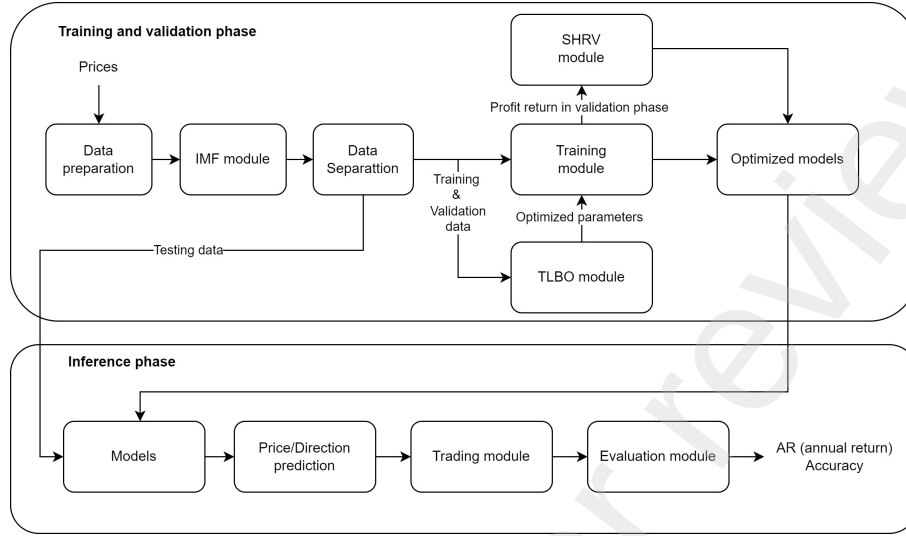


Figure 1: Proposed architectural framework

$$X_i^{new} = \begin{cases} X_i + r * (X_i - X_j) & , f(X_i) < f(X_j) \\ X_i + r * (X_i - X_j) & , otherwise \end{cases} \quad (2)$$

In the third phase, the model clips the parameters to ensure the new values remain within the minimum and maximum values defined in the search space using the formula 3.

$$X_i^{new} = clip(X_i^{new}, min, max) \quad (3)$$

3.4. Proposed framework

This study proposed an architecture to find the optimized models to predict the trend of metal prices. In the training and validation phase, the time series of price is separated into the training and validation dataset, and the testing dataset by Data preparation module. The training and validation data is sent to the IMF module to calculate the IMFs. The IMFs are sent to the Training module for training and the TLBO module for finding the optimized parameters using the TLBO algorithm. We have 3 ways to find the optimized models. The first way is using the IMFs as input data for the Training module to train and validate models only. The second way is applying the TLBO module to calculate the optimized learning rate, training ratio, and validation ratio to train and validate models by the Training module. The third way is applying

the SHRV module to select the model that achieves the highest profit return for the validation dataset provided by the Training module.

In the inference phase, the optimized models are used to predict the price and direction for the testing dataset. The predictions are used by the Trading module to make decisions. Finally, we used the Evaluation module to calculate the Annual return and Accuracy for the testing dataset. The classification system consists of three distinct classes S_i : 1, 0, and -1, corresponding to the respective actions of Buy, No Action, and Sell. S_i is calculated using the formulas 4, where C_i is the close price at the day i , and V_T is the price volatility threshold which is set to 0.1%.

We apply a simple trading strategy for a trading simulation, following the steps below. Firstly, we use the prediction of the next day to make a trading decision. We have three types of decisions including Buy, Sell, and No Action. Secondly, we enter the market at the open day and exit at the end of the day. In the trading strategy, we buy or sell with the same initial balance, so the daily returns DR_i of the day i th can be calculated using the formula 5 with the close and open price of the next day, where the action is -1 for Sell, 1 for Buy, and 0 for No Action. Finally, the total daily return R of T_s trading days is calculated by the sum of all daily returns using the formulas 6 and the average Daily Return (aDR) is calculated using the formula (7).

$$S_{i+1} = \begin{cases} 1 & , C_{i+1} - C_i > V_T \\ -1 & , C_{i+1} - C_i < -V_T \\ 0 & , otherwise \end{cases} \quad (4)$$

$$DR_i = \frac{Close(i) - Open(i)}{Close(i)} * action \quad (5)$$

$$R = \sum_{t=1}^{T_s} DR_i \quad (6)$$

$$aDR = \frac{1}{T_s} \sum_{t=1}^{T_s} DR_i \quad (7)$$

3.5. Formulas and evaluation metrics

The compound return (CR) is calculated by the formula (8), which is the profit earned in a number of trading days T_s , and the number of trading days per year T_y . The annual return (AR) is calculated using the formula (9), where $Balance_{T_0}$ is the initial balance and $Balance_T$ is the balance at the end of the simulation in T_s trading days. The accuracy (Acc) is used to evaluate the price trend prediction, calculated according to formula (10), where P_i is 1 if

the prediction at the day t is correct, and 0 if the prediction is wrong. The simple trading strategy can be described in algorithm 1. Where the balance is calculated for each trading day with actions are Buy or Sell, and the Compound return (CR) is calculated at the end.

$$CR = \frac{Balance_T}{Balance_{T_0}} - 1 \quad (8)$$

$$AR = (1 + CR)^{T_y/T_s} - 1 \quad (9)$$

$$Acc = \frac{1}{T_s} \sum_{t=1}^{T_s} P_i \quad (10)$$

Algorithm 1 Simple trading strategy

```

1: balance  $\leftarrow$  initial-balance
2: for  $i$  in 1 to  $n$ -trading-days do
3:   profit[ $i$ ]  $\leftarrow$  (close-prices[ $i$ ] - open-prices[ $i$ ]) / open-prices[ $i$ ]
4:   if action is "BUY" at day  $i$  then
5:     balance  $\leftarrow$  balance  $\times$  (1 + profit[ $i$ ])
6:   end if
7:   if action is "SELL" at day  $i$  then
8:     balance  $\leftarrow$  balance  $\times$  (1 - profit[ $i$ ])
9:   end if
10: end for
11: CR = balance / initial-balance - 1

```

4. Experiment results and analysis

4.1. Data descriptions and experimental environment

To evaluate the effectiveness of the models, we used the public daily price data of copper, aluminum, nickel, zinc, iron, silver, and gold from investing.com. Some datasets were downloaded with different start and end dates than other datasets. Fig. 2 describe the historical data of 7 metals in the experiments. We separate the datasets into 80% for training and validation and 20% of the last for testing. Normally, the training and validation datasets are separated into 80% for training and 20% of the last for validation. The training and validation datasets are used for the TLBO algorithm to find the optimized training and validation ratio.

Table 1: Model description

Model Number	Model Name	Description
M1	ANN regression	ANN model with loss function is MSE
M2	ANN classification	ANN model with loss function is Cross Entropy
M3	Linear regression	Linear regression model with loss function is MSE
M4	Logistic classification	Logistic classification model with loss function is Cross Entropy
M5	LSTM regression	LSTM model with loss function is MSE
M6	LSTM classification	LSTM model with loss function is Cross Entropy
M7	XGboost regression	XGboost model with loss function is MSE
M8	XGboost classification TLBO	XGboost model with loss function is Cross Entropy
M9	ANN regression TLBO	Apply the TLBO algorithm to find the best learning rate, train ratio, and validation ratio for training and validating the M1, M2, M3, M4, M5, M6, M7, and M8 models respectively. The selected model achieves the lowest loss value for the validation dataset.
M10	ANN Classification TLBO	
M11	Linear regression TLBO	
M12	Logistic classification TLBO	
M13	LSTM regression TLBO	
M14	LSTM classification TLBO	
M15	XGboost regression TLBO	
M16	XGboost classification SHRV	
M17	ANN regression SHRV	The selected model for the prediction phase is the model that achieves the highest profit return for the validation dataset.
M18	ANN Classification SHRV	
M19	Logistic classification SHRV	
M20	LSTM classification TLBO SHRV	
M21	ANN regression TLBO SHRV	Apply the TLBO algorithm to find the best learning rate, train ratio, and validation ratio for training and validating the M9, M10, M12, and M14 models respectively. The selected model for the prediction phase is the model that achieves the highest profit return for the validation dataset.
M22	ANN Classification TLBO SHRV	
M23	Logistic classification TLBO SHRV	
M24	LSTM classification	

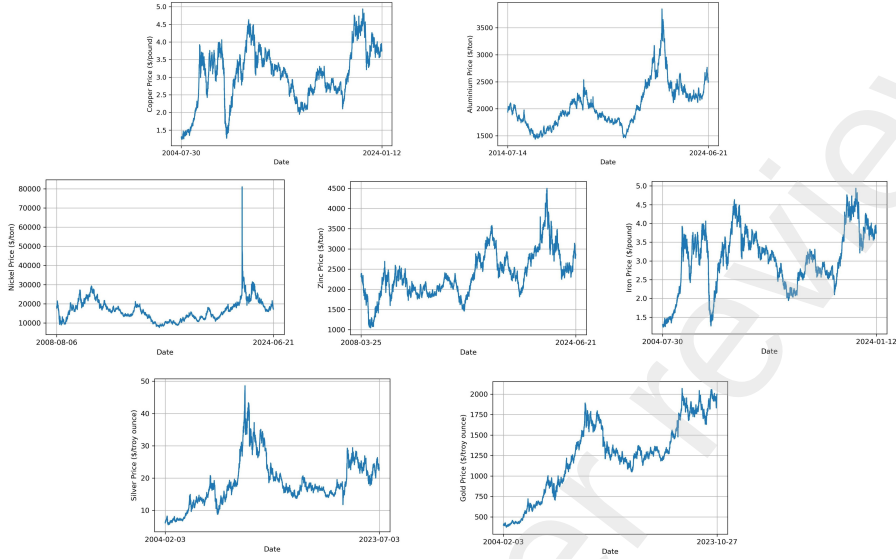


Figure 2: The historical prices of 7 metals in the original dataset.

4.2. Model description

Table 1 describes all models used in our experiments. We train and evaluate the model in 3 different experiments. In experiment 1, we train the models from M1 to M8 with the time series of close price as input, the loss function is MSE for regression models or cross-entropy for classification models. In experiment 2, we train models from M1 to M16 with the IMFs as input and apply the TLBO algorithm to find the optimized learning rate, training, and validation ratio. In experiment 3, we continue to use IMFs as input for training and validating models from M17 to M24, and apply the SHRv module to select the model with the highest annual return in the validation phase for the prediction phase. The TLBO algorithm is applied for models from M21 to M24.

4.3. Experiment results and evaluation

In experiment 1, the input comprises the time series of closing prices for seven metals, including copper, aluminum, nickel, zinc, iron, silver, and gold. We calculate the annual return, average daily return, and accuracy of eight models: ANN regression, ANN classification, Linear regression, Logistic classification, LSTM regression, LSTM classification, XGboost regression, and XGboost classification. In table 2, the ANN regression model achieved the highest annual return of 29.4% and highest average daily return of 0.1174% for Nickel, while the XGboost classification model achieved the highest accuracy with 51.21% for

Table 2: Annual return (AR), average daily return (aDR), and accuracy (Acc) of models in Experiment 1 with time series of closing price as input.

Model	Metric (%)	7 types of metals							Average (%)
		Copper	Alu	Nickel	Zinc	Iron	Silver	Gold	
M1	AR	15.52	25.23	29.40	-1.23	15.52	-32.83	-4.08	6.79
	aDR	0.0667	0.0933	0.1174	0.0004	0.0667	- 0.1367	- 0.0116	0.0280
	Acc	46.93	49.11	49.30	45.82	46.93	43.62	43.12	46.40
M2	AR	-10.89	-6.99	-11.91	-23.85	-3.27	-22.71	2.87	-10.96
	aDR	-0.0362	- 0.0247	- 0.0349	- 0.1028	- 0.0038	- 0.0807	0.0161	-0.0381
	Acc	45.53	48.91	47.64	43.57	45.53	47.64	42.31	45.88
M3	AR	15.33	26.69	27.32	-1.23	15.33	-32.83	-3.97	6.66
	aDR	0.0661	0.0979	0.1109	0.0004	0.0661	- 0.1367	- 0.0112	0.0276
	Acc	46.83	49.30	49.17	45.82	46.83	43.62	43.22	46.40
M4	AR	-2.43	-11.89	27.46	2.39	-1.69	-16.68	8.73	0.84
	aDR	-0.0002	- 0.0462	0.1117	0.0147	0.0028	- 0.0510	0.0381	0.0100
	Acc	48.54	48.31	48.02	46.19	48.74	49.25	42.91	47.42
M5	AR	13.96	-11.89	28.70	12.61	13.96	-16.68	0.24	5.84
	aDR	0.0614	- 0.0462	0.1155	0.0525	0.0614	-0.051	0.0059	0.0285
	Acc	48.44	48.31	47.51	48.19	48.44	49.25	40.70	47.26
M6	AR	14.01	-11.89	28.70	12.61	-6.15	-3.46	8.99	6.12
	aDR	0.0616	- 0.0462	0.1155	0.0525	- 0.0156	0.0073	0.0391	0.0306
	Acc	49.75	48.31	47.51	48.19	46.93	49.65	44.02	47.77
M7	AR	13.96	-3.26	16.95	-0.34	13.96	-13.32	3.44	4.48
	aDR	0.0614	- 0.0091	0.0775	0.0040	0.0614	- 0.0353	0.0183	0.0255
	Acc	48.44	49.30	49.68	45.32	48.44	50.05	42.21	47.63
M8	AR	2.70	-10.18	2.74	3.00	2.70	-17.62	17.03	0.05
	aDR	0.0201	- 0.0386	0.0261	0.0171	0.0201	- 0.0555	0.0673	0.0081
	Acc	47.34	48.71	51.21	45.32	47.34	48.04	45.13	47.58

Nickel. Upon calculating the average performance of all models, it is evident that ANN regression achieves the highest annual return of 6.79%, while the LSTM classification model achieves the highest average daily return of 0.0306% and highest accuracy of 47.77%. The experiment demonstrates that utilizing closing prices as input yields low performance in annual and daily returns, with numerous models producing negative values.

In experiment 2, we calculated Intrinsic Mode Functions (IMFs) for each time series of close prices to utilize as input for models. The results in Table 3 demonstrate that we have significantly improved performance when employing IMFs as features for all models including M1, M2, M3, M4, M5, M6, M7, and M8. Table 3 indicates that these models exhibit significant improvements, achieving the highest average annual return of 314.32% and the highest average accuracy of 68.38% with the M8 model using XGboost classification.

Subsequently, we applied the Teaching-Learning-Based Optimization (TLBO) algorithm to enhance these models in the training and validation phase by determining the optimized learning rate, training ratio, and validation ratio. Consequently, we developed M9, M10, M11, M12, M13, M14, M15, and M16 models, which incorporate the optimized parameters from the TLBO modules. In experiment 2, the results indicate that TLBO provided optimized learning rate, training ratio, and validation ratio to improve the performance of all classification-based models including ANN Classification, Logistic classification, LSTM classification, and XGboost classification. With these optimized parameters, we increased the annual return of M2, M4, M6, M8 on average. The M12 model using Logistic classification achieved the highest improvement in annual return of 12.4704%. This demonstrates that TLBO can identify more optimal parameters for classification-based models compared to regression-based models.

The results in Table 3 indicate that we achieved the highest annual return of 829.9% with the M10 model using ANN Classification for Silver, and the highest accuracy of 72.91% with the M6 model using LSTM classification for Zinc. In the average calculation for all types of metals, we achieved the highest annual return of 330.61% and the highest average accuracy of 69.15% with the M16 model using XGboost classification optimized by the TLBO algorithm. This provides evidence that TLBO can enhance model predictions to increase profit return.

In experiment 3, we employed the SHRV module to select models that achieve the highest profit return instead of the lowest loss in the validation phase to predict the price trend in the inference phase. We utilized the optimized learning rate, training ratio, and validation ratio for ANN regression, ANN Classification, Logistic classification, and LSTM classification models from experiment 2 to train and validate the M21, M22, M23, and M24 models respectively.

Table 3: Annual return, average daily return, and accuracy of Models in Experiment 1 with time series of closing price as input.

Model	Metric (%)	7 types of metals							Average (%)	TLBO improvement
		Copper	Alu	Nickel	Zinc	Iron	Silver	Gold		
M1	AR	230.80	75.70	200.24	98.29	230.96	371.55	127.60	190.73	-
	aDR	0.4842	0.2277	0.4512	0.2770	0.4844	0.6361	0.3312	0.4131	-
	Acc	59.70	60.44	62.58	59.80	59.80	59.60	58.09	60.00	-
M9	AR	237.41	77.05	207.36	97.24	198.19	417.99	98.53	190.54	-0.0996%
	aDR	0.4920	0.2307	0.4605	0.2749	0.4430	0.6734	0.2770	0.4074	-1.3798%
	Acc	60.20	61.03	63.09	59.68	57.79	60.30	54.57	59.52	-0.8000%
M2	AR	330.78	74.61	110.73	111.71	384.07	802.97	206.18	288.72	-
	aDR	0.5890	0.2252	0.3107	0.3030	0.6352	0.8938	0.4489	0.4865	-
	Acc	66.33	62.82	64.50	69.91	70.05	68.44	64.82	66.70	-
M10	AR	394.65	83.73	108.78	121.80	402.05	829.90	223.17	309.15	7.0761%
	aDR	0.6438	0.2454	0.3071	0.3214	0.6497	0.9054	0.4703	0.5062	4.0493%
	Acc	70.05	66.60	56.19	72.41	70.25	69.75	66.83	67.44	1.1094%
M3	AR	253.13	28.56	45.30	-16.90	253.13	402.52	-31.38	133.48	-
	aDR	0.5101	0.1037	0.1634	-	0.5101	0.6616	-	0.2480	-
	Acc	59.70	49.90	52.62	42.45	59.70	61.91	38.79	52.15	-
M11	AR	250.20	28.23	45.61	-14.59	253.13	389.31	-25.84	132.29	-0.8915%
	aDR	0.5068	0.1027	0.1642	-	0.5101	0.6508	-	0.2520	1.6129%
	Acc	59.60	49.70	52.75	42.57	59.70	60.10	40.00	52.06	-0.1726%
M4	AR	402.52	77.57	96.68	111.63	402.52	784.47	222.28	299.67	-
	aDR	0.6501	0.2319	0.2834	0.3028	0.6501	0.8856	0.4693	0.4962	-
	Acc	70.75	66.00	54.79	69.66	70.75	69.95	66.73	66.95	-
M12	AR	396.06	72.85	265.38	128.53	400.24	921.49	174.71	337.04	12.4704%
	aDR	0.6449	0.2212	0.5290	0.3333	0.6483	0.9427	0.4059	0.5322	7.2551%
	Acc	70.55	65.21	67.69	72.41	70.15	70.85	63.62	68.64	2.5243%
M6	AR	270.42	82.12	9.12	119.27	378.43	829.30	229.04	273.96	-
	aDR	0.5291	0.2419	0.0497	0.3169	0.6306	0.9052	0.4775	0.4501	-
	Acc	64.62	64.61	51.60	72.91	69.35	69.15	66.63	65.55	-
M14	AR	405.17	71.12	98.76	65.88	271.41	866.19	219.47	285.43	4.1867%
	aDR	0.6522	0.2171	0.2866	0.2062	0.5301	0.9206	0.4657	0.4684	4.0658%
	Acc	69.75	60.24	55.56	60.55	63.12	70.35	65.63	63.60	-2.9748%
M7	AR	31.23	18.18	3.54	9.41	31.23	44.32	8.80	20.96	-
	aDR	0.1110	0.0687	0.0226	0.0388	0.1110	0.1523	0.0345	0.0770	-
	Acc	22.11	33.20	30.40	29.46	22.11	18.59	19.60	25.07	-
M15	AR	0.68	20.24	-9.21	-15.11	-2.01	-30.63	-17.09	-7.59	-136.2118%
	aDR	0.0121	0.0772	-	-	0.0014	-0.124	-	-0.0267	-134.6753%
	Acc	43.92	44.53	45.72	41.95	44.42	44.02	40.90	43.64	74.0726%
M8	AR	375.46	72.20	260.85	121.31	375.46	802.93	192.06	314.32	-
	aDR	0.6281	0.2197	0.5236	0.3205	0.6281	0.8938	0.4302	0.5206	-
	Acc	69.55	67.00	67.82	71.16	69.55	69.65	63.92	68.38	-
M16	AR	378.80	85.92	307.98	114.71	339.67	876.52	210.67	330.61	5.1826%
	aDR	0.6309	0.2501	0.5728	0.3085	0.5971	0.9249	0.4547	0.5341	2.5932
	Acc	69.45	68.59	71.52	70.29	68.44	70.95	64.82	69.15	1.1261

Table 4: Annual return, average daily returns, and accuracy of models in Experiment 3 with IMF as Input. The experiment used the SHRV module to select the model that achieved the highest annual return in the validation phase for the inference phase. Experiment applied TLBO algorithm and evaluate the TLBO improvement of performance.

Model	Metric (%)	7 types of metals							Average (%)	TLBO improvement
		Copper	Alu	Nickel	Zinc	Iron	Silver	Gold		
M17	AR	288.60	75.70	193.46	88.56	196.74	461.99	145.01	207.15	-
	aDR	0.5481	0.2277	0.4422	0.2570	0.4411	0.7060	0.3605	0.4261	-
	Acc	61.21	60.44	62.45	65.42	59.20	63.22	59.30	61.61	-
M21	AR	231.34	75.92	215.82	102.06	227.79	420.16	144.98	202.58	-2.2061%
	aDR	0.4848	0.2282	0.4713	0.2844	0.4806	0.6750	0.3604	0.4264	0.0704%
	Acc	58.99	60.44	63.60	60.67	59.10	61.01	59.50	60.47	-1.8503%
M18	AR	324.47	80.17	125.26	129.34	360.74	544.27	221.98	255.18	-
	aDR	0.5831	0.2376	0.3372	0.3347	0.6156	0.7587	0.4689	0.4765	-
	Acc	66.23	67.99	58.62	72.41	68.24	60.30	66.53	65.76	-
M22	AR	413.78	87.45	300.74	117.05	392.36	930.81	221.66	351.98	37.9340%
	aDR	0.6589	0.2533	0.5657	0.3128	0.6420	0.9463	0.4685	0.5496	15.3410%
	Acc	70.25	69.18	71.01	71.66	70.05	71.56	66.33	70.01	6.4629%
M19	AR	221.81	78.23	16.28	53.42	383.20	793.21	224.72	252.98	-
	aDR	0.4732	0.2333	0.0749	0.1752	0.6345	0.8895	0.4722	0.4218	-
	Acc	63.12	67.20	48.79	60.30	69.55	69.95	65.63	63.51	-
M23	AR	404.67	84.15	230.35	126.40	376.14	853.38	223.51	328.37	29.8008%
	aDR	0.6518	0.2463	0.4890	0.3296	0.6287	0.9153	0.4708	0.5331	26.3869%
	Acc	70.15	67.00	66.28	72.66	69.25	70.85	67.04	69.03	8.6915%
M20	AR	292.99	78.92	234.16	112.49	379.06	889.49	220.12	315.32	-
	aDR	0.5525	0.2349	0.4937	0.3044	0.6311	0.9301	0.4666	0.5162	-
	Acc	65.43	65.21	67.05	72.53	69.65	71.26	66.13	68.18	-
M24	AR	397.65	80.90	271.97	125.15	361.35	645.38	222.59	300.71	-4.6334%
	aDR	0.6462	0.2392	0.5362	0.3274	0.6155	0.8173	0.4696	0.5216	1.0461%
	Acc	70.25	68.79	68.20	71.79	68.24	68.14	65.63	68.72	0.7920%

Table 5: Compare the performance of the best models in all experiments.

Model	Algorithm	Average performance			TLBO improvement		
		Annual return (%)	Average daily return (%)	Accuracy (%)	Annual return (%)	Average daily return (%)	Accuracy (%)
M9	ANN regression	190.54	0.4074	59.52	-0.0996	-1.3798	-0.8000
M21		202.58	0.4264	60.47	-2.2061	0.0704	-1.8503
M10	ANN classification	309.15	0.5062	67.44	7.0761	4.0493	1.1094
M22		351.98	0.5496	70.01	37.934	15.341	6.4629
M12	Logistic classification	337.04	0.5322	68.64	12.4704	7.2551	2.5243
M23		328.37	0.5331	69.03	29.8008	26.3869	8.6915
M14	LSTM classification	285.43	0.4684	63.60	4.1867	4.0658	-2.9748
M24		300.71	0.5216	68.72	-4.6334	1.0461	0.7920

The results presented in Table 4 indicate that the M22 model utilizing ANN Classification attains the highest TLBO improvement with an average annual return of 37.934%, while the M23 model employing Logistic classification achieves the highest improvement in average accuracy of 8.6915%.

In experiment 3, the M22 model utilizing ANN Classification attains the highest annual return of 930.81% for silver, and the M23 model employing Logistic classification achieves the highest accuracy of 72.66% for zinc. The lowest annual return of 16.28% for aluminum is associated with the M19 Model utilizing Logistic classification. In the average calculation across all metal types, the M22 model attains the highest average annual return of 351.98% and the highest average accuracy of 70.01%. All models in the experiment achieve an average annual return exceeding 54%.

Table 5 presents a comparison of the performance of the optimal models across all experiments. The results indicate that the application of the TLBO algorithm leads to an increase in average daily returns for all classification-based models, including ANN Classification, Logistic classification, and LSTM classification. For instance, the M23 model utilizing Logistic classification demonstrates a TLBO improvement of 26.3869%, while the M22 model employing ANN classification exhibits a TLBO improvement of 15.341% in average daily returns. The implementation of the SHRV module results in improved annual returns for algorithms including ANN regression, ANN Classification, and LSTM classification. As an example, the M22 model achieves an annual return

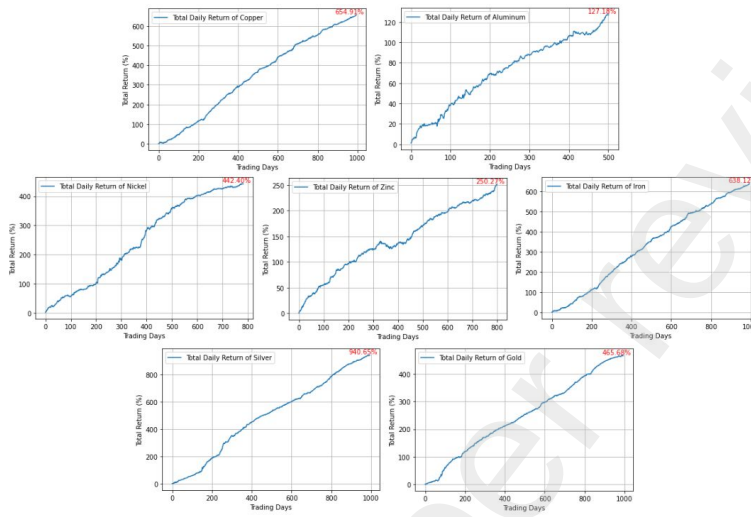


Figure 3: The total daily returns in the inference phase of the M22 model.

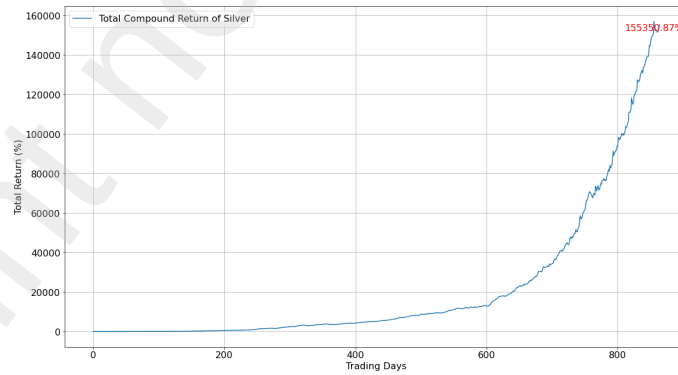


Figure 4: The total compound return for silver in the inference phase of the M22 model.

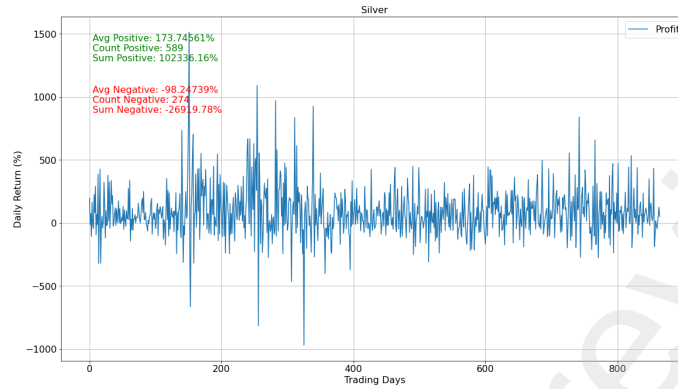


Figure 5: The daily profit for silver in the inference phase of the M22 model.

of 351.98%, which surpasses the M10 model's 309.15%, while the M21 model attains an annual return of 202.58%, exceeding the M10 model's 190.54%.

Across all experiments, the M22 model, optimized by TLBO and incorporating SHRV, achieved the highest average annual return, daily return, and accuracy of 351.98%, 0.5496%, and 70.01%, respectively. These findings demonstrate that the application of TLBO and SHRV algorithms enhances the performance of classification-based models in generating increased profits within trading simulations.

4.4. Evaluate the return of the best performance model

Fig. 3 illustrates the total daily return of the M22 model during the inference phase, utilizing the testing dataset comprising seven metal types: copper, aluminum, nickel, zinc, iron, silver, and gold. The results demonstrate that the total returns of the M22 model exhibit an increasing trend over time when employing our straightforward trading strategy. In the experimental phase, we observed total daily returns of 654.91% for copper, 638.12% for iron, 465.68% for gold, and a maximum of 940.65% for silver over a period of 995 trading days.

Fig. 4 illustrates the total compound return of silver when utilizing algorithm 1 to calculate the final balance after 995 trading days. The results demonstrate a substantial increase in balance by a factor of 1553. Fig. 5 depicts the daily profit in the experiment, revealing 589 profitable trades with positive returns and 274 unprofitable trades with negative returns. Notably, the average positive return of 1.72746% is approximately 1.76 times greater than the average negative return of -0.98247%. This outcome indicates that the M22 model not only generates profitable trades more frequently than unprofitable ones, but also that the magnitude of profit for successful trades exceeds the magnitude of loss for unsuccessful trades. These findings suggest that the proposed method can provide reliable predictions for generating profit.

Table 6: Compare the performance of the best models in all experiments.

Paper	Metal	Model	Date	Accuracy (%)	Return (%)
Proposed method	Copper	M22	2017/7/7-	70	14.71
Wang et al. [15]		PVN-ELM		2017/8/17	53.33
		PVN-BP	56.67		-
		PVN-RBF	60		-

4.5. Performance comparison

Table 6 presents a comparative analysis between the proposed method and other studies. The results of the present study were compared with those of Wang et al. [15]. Their study proposed a hybrid technique for copper price forecasting utilizing Artificial Neural Network (ANN) algorithms, including Back Propagation Neural Network (BPNN), Radial Basis Function Neural Network (RBFNN), and Extreme Learning Machine (ELM), combined with a transformation method called Price Volatility Network (PVN). For daily prediction, their study achieved optimal performance with PVN-ELM, PVN-BP, and PVN-RBF, yielding accuracies of 53.33%, 56.67%, and 60%, respectively. However, their study did not implement any trading strategy to calculate the return on their predictions. In contrast, the present study's proposed M22 model, utilizing the ANN classifier, achieved a higher accuracy of 70% in copper price prediction and generated a return of 14.71% over 30 trading days.

5. Conclusion and Future Works

In this study, we propose a novel framework to predict the trends of metal prices. We conduct experiments with machine-learning algorithms, including linear regression, logistic regression, Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), and XGBoost, for both classification and regression output. The results indicate that the classification-based model using Intrinsic Mode Functions (IMFs) as input features, and optimized by Teaching-Learning-Based Optimization (TLBO) and Selecting highest return at validation (SHRV) based methods demonstrated superior performance compared to other models. In our experimental analysis, the M22 model utilizing ANN classification and optimized by both TLBO and SHRV achieved the highest performance with an average annual return of 351.98%, a daily return of 0.5496%, and an accuracy of 70.01% for seven types of metals. The proposed method exhibits high accuracy in predicting significant price movements, which is a crucial aspect of price prediction. Utilizing IMFs as input features can significantly enhance performance, with all classification-based models achieving an average annual return exceeding 273.96%. The proposed method demonstrates high accuracy in predicting significant price movements, which is a critical aspect of price prediction.

In future work, we intend to expand the model's applicability to incorporate additional features, including specific technical indicators and text-based features derived from news sources or messaging platforms. Furthermore, we aim to evaluate the model using various time frames, such as weekly and monthly intervals. The model is designed to process diverse stock and commodity price data; therefore, we also intend to apply the proposed models to predict the prices, volatility, and trends of other metals, indices, and foreign exchange markets. Lastly, we plan to develop methodologies to integrate optimized models trained on different time frames into a real-time system, with the objective of maximizing returns.

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